Lab 5

```
from google.colab import files

uploaded = files.upload()
```

```
   <IPython.core.display.HTML object>


   Saving food_delivery_orders_dataset.csv to food_delivery_orders_dataset.csv
```

```
import pandas as pd

filename = next(iter(uploaded))
df = pd.read_csv(filename)

df.head()
```

```
       order_id customer_id restaurant_id driver_id      order_timestamp  \
   0  ORD_000001   CUS_00001      RES_0008  DRV_0075  2025-10-05 15:54:00
   1  ORD_000002   CUS_00013      RES_0024  DRV_0131  2025-10-01 20:00:00
   2  ORD_000003   CUS_00049      RES_0013  DRV_0087  2025-10-07 18:24:00
   3  ORD_000004   CUS_00007      RES_0001  DRV_0177  2025-10-30 16:54:00
   4  ORD_000005   CUS_00325      RES_0029  DRV_0114  2025-10-15 18:34:00

      order_date  order_hour day_of_week  is_weekend    city  ... traffic_level  \
   0  2025-10-05          15      Sunday           1  City_C  ...        Medium
   1  2025-10-01          20   Wednesday           0  City_A  ...          High
   2  2025-10-07          18     Tuesday           0  City_C  ...          High
   3  2025-10-30          16    Thursday           0  City_A  ...           Low
   4  2025-10-15          18   Wednesday           0  City_B  ...        Medium

      delivery_partner_experience_months delivery_partner_rating  \
   0                                  14                     4.4
   1                                  44                     4.7
   2                                   1                     5.0
   3                                  50                     4.3
   4                                  21                     4.0

     restaurant_preparation_time_minutes estimated_delivery_time_minutes  \
   0                                  25                              25
   1                                  12                              20
   2                                  11                              63
   3                                  12                              23
   4                                   9                              32

      actual_delivery_time_minutes  late_delivery  order_status  \
   0                          29.0            0.0     Completed
   1                          24.0            0.0     Completed
   2                          72.0            1.0     Completed
   3                          31.0            1.0     Completed
   4                          42.0            1.0     Completed

      cancellation_reason  customer_rating
   0                  NaN              4.6
   1                  NaN              5.0
   2                  NaN              3.6
   3                  NaN              4.1
   4                  NaN              3.9
```

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```
[5 rows x 37 columns]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 37 columns):
 #   Column                                Non-Null Count  Dtype  
---  ------                                --------------  -----  
 0   order_id                              50000 non-null  object 
 1   customer_id                           50000 non-null  object 
 2   restaurant_id                         50000 non-null  object 
 3   driver_id                             50000 non-null  object 
 4   order_timestamp                       50000 non-null  object 
 5   order_date                            50000 non-null  object 
 6   order_hour                            50000 non-null  int64  
 7   day_of_week                           50000 non-null  object 
 8   is_weekend                            50000 non-null  int64  
 9   city                                  50000 non-null  object 
 10  delivery_area                         50000 non-null  object 
 11  customer_age                          50000 non-null  int64  
 12  customer_type                         50000 non-null  object 
 13  restaurant_type                       50000 non-null  object 
 14  restaurant_primary_category           50000 non-null  object 
 15  restaurant_rating                     50000 non-null  float64
 16  items_count                           50000 non-null  int64  
 17  subtotal                              50000 non-null  float64
 18  discount_percent                      50000 non-null  int64  
 19  tax_amount                            50000 non-null  float64
 20  service_fee                           50000 non-null  float64
 21  delivery_fee                          50000 non-null  float64
 22  order_total                           50000 non-null  float64
 23  payment_method                        50000 non-null  object 
 24  tip_amount                            49132 non-null  float64
 25  distance_km                           50000 non-null  float64
 26  weather                               50000 non-null  object 
 27  traffic_level                         50000 non-null  object 
 28  delivery_partner_experience_months    50000 non-null  int64  
 29  delivery_partner_rating               50000 non-null  float64
 30  restaurant_preparation_time_minutes   50000 non-null  int64  
 31  estimated_delivery_time_minutes       50000 non-null  int64  
 32  actual_delivery_time_minutes          49132 non-null  float64
 33  late_delivery                         49132 non-null  float64
 34  order_status                          50000 non-null  object 
 35  cancellation_reason                   868 non-null    object 
 36  customer_rating                       49132 non-null  float64
dtypes: float64(12), int64(8), object(17)
memory usage: 14.1+ MB
```

```
df.columns
```

```
Index(['order_id', 'customer_id', 'restaurant_id', 'driver_id',
       'order_timestamp', 'order_date', 'order_hour', 'day_of_week',
       'is_weekend', 'city', 'delivery_area', 'customer_age', 'customer_type',
       'restaurant_type', 'restaurant_primary_category', 'restaurant_rating',
       'items_count', 'subtotal', 'discount_percent', 'tax_amount',
       'service_fee', 'delivery_fee', 'order_total', 'payment_method',
       'tip_amount', 'distance_km', 'weather', 'traffic_level',
       'delivery_partner_experience_months', 'delivery_partner_rating',
```

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```
       'restaurant_preparation_time_minutes',
       'estimated_delivery_time_minutes', 'actual_delivery_time_minutes',
       'late_delivery', 'order_status', 'cancellation_reason',
       'customer_rating'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 37 columns):
 #   Column                                Non-Null Count  Dtype  
---  ------                                --------------  -----  
 0   order_id                              50000 non-null  object 
 1   customer_id                           50000 non-null  object 
 2   restaurant_id                         50000 non-null  object 
 3   driver_id                             50000 non-null  object 
 4   order_timestamp                       50000 non-null  object 
 5   order_date                            50000 non-null  object 
 6   order_hour                            50000 non-null  int64  
 7   day_of_week                           50000 non-null  object 
 8   is_weekend                            50000 non-null  int64  
 9   city                                  50000 non-null  object 
 10  delivery_area                         50000 non-null  object 
 11  customer_age                          50000 non-null  int64  
 12  customer_type                         50000 non-null  object 
 13  restaurant_type                       50000 non-null  object 
 14  restaurant_primary_category           50000 non-null  object 
 15  restaurant_rating                     50000 non-null  float64
 16  items_count                           50000 non-null  int64  
 17  subtotal                              50000 non-null  float64
 18  discount_percent                      50000 non-null  int64  
 19  tax_amount                            50000 non-null  float64
 20  service_fee                           50000 non-null  float64
 21  delivery_fee                          50000 non-null  float64
 22  order_total                           50000 non-null  float64
 23  payment_method                        50000 non-null  object 
 24  tip_amount                            49132 non-null  float64
 25  distance_km                           50000 non-null  float64
 26  weather                               50000 non-null  object 
 27  traffic_level                         50000 non-null  object 
 28  delivery_partner_experience_months    50000 non-null  int64  
 29  delivery_partner_rating               50000 non-null  float64
 30  restaurant_preparation_time_minutes   50000 non-null  int64  
 31  estimated_delivery_time_minutes       50000 non-null  int64  
 32  actual_delivery_time_minutes          49132 non-null  float64
 33  late_delivery                         49132 non-null  float64
 34  order_status                          50000 non-null  object 
 35  cancellation_reason                   868 non-null    object 
 36  customer_rating                       49132 non-null  float64
dtypes: float64(12), int64(8), object(17)
memory usage: 14.1+ MB
```

```
df.describe(include='all')
```

```
           order_id customer_id restaurant_id driver_id      order_timestamp  \
count        50000       50000         50000     50000                50000   
unique       50000        4979           250       200                25459   
top     ORD_049984   CUS_00001      RES_0001  DRV_0066  2025-10-11 19:51:00   
freq             1        4510          8837       315                    9   
```

mean	NaN	NaN	NaN	NaN		NaN
std	NaN	NaN	NaN	NaN		NaN
min	NaN	NaN	NaN	NaN		NaN
25%	NaN	NaN	NaN	NaN		NaN
50%	NaN	NaN	NaN	NaN		NaN
75%	NaN	NaN	NaN	NaN		NaN
max	NaN	NaN	NaN	NaN		NaN

	order_date	order_hour	day_of_week	is_weekend	city	...	\
count	50000	50000.000000	50000	50000.00000	50000	...	
unique	30	NaN	7	NaN	4	...	
top	2025-10-23	NaN	Thursday	NaN	City_A	...	
freq	1741	NaN	8472	NaN	19888	...	
mean	NaN	14.465900	NaN	0.26712	NaN	...	
std	NaN	5.231155	NaN	0.44246	NaN	...	
min	NaN	0.000000	NaN	0.00000	NaN	...	
25%	NaN	11.000000	NaN	0.00000	NaN	...	
50%	NaN	14.000000	NaN	0.00000	NaN	...	
75%	NaN	19.000000	NaN	1.00000	NaN	...	
max	NaN	23.000000	NaN	1.00000	NaN	...	

	traffic_level	delivery_partner_experience_months	\
count	50000	50000.0000	
unique	3	NaN	
top	Medium	NaN	
freq	19522	NaN	
mean	NaN	29.7332	
std	NaN	17.4052	
min	NaN	1.0000	
25%	NaN	14.0000	
50%	NaN	29.0000	
75%	NaN	44.0000	
max	NaN	60.0000	

	delivery_partner_rating	restaurant_preparation_time_minutes	\
count	50000.000000	50000.000000	
unique	NaN	NaN	
top	NaN	NaN	
freq	NaN	NaN	
mean	4.495062	15.549600	
std	0.349198	7.360099	
min	3.200000	5.000000	
25%	4.300000	10.000000	
50%	4.500000	14.000000	
75%	4.800000	19.000000	
max	5.000000	73.000000	

	estimated_delivery_time_minutes	actual_delivery_time_minutes	\
count	50000.000000	49132.000000	
unique	NaN	NaN	
top	NaN	NaN	
freq	NaN	NaN	
mean	30.950580	35.152365	
std	12.825519	14.701154	
min	8.000000	10.000000	
25%	22.000000	25.000000	
50%	28.000000	33.000000	
75%	37.000000	42.000000	
max	128.000000	172.000000	

	late_delivery	order_status	cancellation_reason	customer_rating
count	49132.000000	50000	868	49132.000000
unique	NaN	2	5	NaN
top	NaN	Completed	Driver Unassigned	NaN

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freq	NaN	49132	386	NaN	
mean	0.365709	NaN	NaN	4.295946	
std	0.481633	NaN	NaN	0.530947	
min	0.000000	NaN	NaN	1.800000	
25%	0.000000	NaN	NaN	4.000000	
50%	0.000000	NaN	NaN	4.400000	
75%	1.000000	NaN	NaN	4.700000	
max	1.000000	NaN	NaN	5.000000	

[11 rows x 37 columns]

```
df.isnull().sum()
```

```
order_id                                0
customer_id                             0
restaurant_id                           0
driver_id                               0
order_timestamp                         0
order_date                              0
order_hour                              0
day_of_week                             0
is_weekend                              0
city                                    0
delivery_area                           0
customer_age                            0
customer_type                           0
restaurant_type                         0
restaurant_primary_category             0
restaurant_rating                       0
items_count                             0
subtotal                                0
discount_percent                        0
tax_amount                              0
service_fee                             0
delivery_fee                            0
order_total                             0
payment_method                          0
tip_amount                            868
distance_km                             0
weather                                 0
traffic_level                           0
delivery_partner_experience_months      0
delivery_partner_rating                 0
restaurant_preparation_time_minutes     0
estimated_delivery_time_minutes         0
actual_delivery_time_minutes          868
late_delivery                         868
order_status                            0
cancellation_reason                 49132
customer_rating                       868
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.columns
```

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```
Index(['order_id', 'customer_id', 'restaurant_id', 'driver_id',
       'order_timestamp', 'order_date', 'order_hour', 'day_of_week',
       'is_weekend', 'city', 'delivery_area', 'customer_age', 'customer_type',
       'restaurant_type', 'restaurant_primary_category', 'restaurant_rating',
       'items_count', 'subtotal', 'discount_percent', 'tax_amount',
       'service_fee', 'delivery_fee', 'order_total', 'payment_method',
       'tip_amount', 'distance_km', 'weather', 'traffic_level',
       'delivery_partner_experience_months', 'delivery_partner_rating',
       'restaurant_preparation_time_minutes',
       'estimated_delivery_time_minutes', 'actual_delivery_time_minutes',
       'late_delivery', 'order_status', 'cancellation_reason',
       'customer_rating'],
      dtype='object')
```

```
df.head(10)
```

	order_id	customer_id	restaurant_id	driver_id	order_timestamp	\
0	ORD_000001	CUS_00001	RES_0008	DRV_0075	2025-10-05 15:54:00	
1	ORD_000002	CUS_00013	RES_0024	DRV_0131	2025-10-01 20:00:00	
2	ORD_000003	CUS_00049	RES_0013	DRV_0087	2025-10-07 18:24:00	
3	ORD_000004	CUS_00007	RES_0001	DRV_0177	2025-10-30 16:54:00	
4	ORD_000005	CUS_00325	RES_0029	DRV_0114	2025-10-15 18:34:00	
5	ORD_000006	CUS_00580	RES_0019	DRV_0112	2025-10-23 14:27:00	
6	ORD_000007	CUS_00021	RES_0002	DRV_0109	2025-10-29 19:20:00	
7	ORD_000008	CUS_00003	RES_0044	DRV_0066	2025-10-26 07:22:00	
8	ORD_000009	CUS_00002	RES_0003	DRV_0155	2025-10-24 14:28:00	
9	ORD_000010	CUS_00009	RES_0009	DRV_0126	2025-10-14 20:32:00	

	order_date	order_hour	day_of_week	is_weekend	city	...	traffic_level	\
0	2025-10-05	15	Sunday	1	City_C	...	Medium	
1	2025-10-01	20	Wednesday	0	City_A	...	High	
2	2025-10-07	18	Tuesday	0	City_C	...	High	
3	2025-10-30	16	Thursday	0	City_A	...	Low	
4	2025-10-15	18	Wednesday	0	City_B	...	Medium	
5	2025-10-23	14	Thursday	0	City_B	...	Medium	
6	2025-10-29	19	Wednesday	0	City_D	...	Medium	
7	2025-10-26	7	Sunday	1	City_D	...	Low	
8	2025-10-24	14	Friday	0	City_C	...	High	
9	2025-10-14	20	Tuesday	0	City_B	...	Medium	

	delivery_partner_experience_months	delivery_partner_rating	\
0	14	4.4	
1	44	4.7	
2	1	5.0	
3	50	4.3	
4	21	4.0	
5	34	5.0	
6	43	4.8	
7	53	4.0	
8	46	4.5	
9	47	4.0	

	restaurant_preparation_time_minutes	estimated_delivery_time_minutes	\
0	25	25	
1	12	20	
2	11	63	
3	12	23	
4	9	32	
5	12	32	
6	11	27	
7	12	20	
8	11	37	

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```
9                                 14                                25

   actual_delivery_time_minutes  late_delivery  order_status  \
0                          29.0            0.0     Completed
1                          24.0            0.0     Completed
2                          72.0            1.0     Completed
3                          31.0            1.0     Completed
4                          42.0            1.0     Completed
5                          39.0            1.0     Completed
6                          31.0            0.0     Completed
7                          22.0            0.0     Completed
8                          40.0            0.0     Completed
9                          22.0            0.0     Completed

   cancellation_reason  customer_rating
0                  NaN              4.6
1                  NaN              5.0
2                  NaN              3.6
3                  NaN              4.1
4                  NaN              3.9
5                  NaN              4.0
6                  NaN              4.5
7                  NaN              4.8
8                  NaN              5.0
9                  NaN              4.8

[10 rows x 37 columns]
```

```
df.dtypes
```

```
order_id                               object
customer_id                            object
restaurant_id                          object
driver_id                              object
order_timestamp                        object
order_date                             object
order_hour                              int64
day_of_week                            object
is_weekend                              int64
city                                   object
delivery_area                          object
customer_age                            int64
customer_type                          object
restaurant_type                        object
restaurant_primary_category            object
restaurant_rating                     float64
items_count                             int64
subtotal                              float64
discount_percent                        int64
tax_amount                            float64
service_fee                           float64
delivery_fee                          float64
order_total                           float64
payment_method                         object
tip_amount                            float64
distance_km                           float64
weather                                object
traffic_level                          object
delivery_partner_experience_months      int64
delivery_partner_rating               float64
restaurant_preparation_time_minutes     int64
estimated_delivery_time_minutes         int64
```

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```
actual_delivery_time_minutes          float64
late_delivery                         float64
order_status                           object
cancellation_reason                    object
customer_rating                       float64
dtype: object
```

```
df = df.drop_duplicates()
```

```
df = df.dropna()
```

```
df.shape
```

```
(0, 37)
```

```
df.nunique()
```

```
order_id                              0
customer_id                           0
restaurant_id                         0
driver_id                             0
order_timestamp                       0
order_date                            0
order_hour                            0
day_of_week                           0
is_weekend                            0
city                                  0
delivery_area                         0
customer_age                          0
customer_type                         0
restaurant_type                       0
restaurant_primary_category           0
restaurant_rating                     0
items_count                           0
subtotal                              0
discount_percent                      0
tax_amount                            0
service_fee                           0
delivery_fee                          0
order_total                           0
payment_method                        0
tip_amount                            0
distance_km                           0
weather                               0
traffic_level                         0
delivery_partner_experience_months    0
delivery_partner_rating               0
restaurant_preparation_time_minutes   0
estimated_delivery_time_minutes       0
actual_delivery_time_minutes          0
late_delivery                         0
order_status                          0
cancellation_reason                   0
customer_rating                       0
dtype: int64
```

```
df.describe()
```

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	order_hour	is_weekend	customer_age	restaurant_rating	items_count	\
count	0.0	0.0	0.0	0.0	0.0	
mean	NaN	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	NaN	

	subtotal	discount_percent	tax_amount	service_fee	delivery_fee	\
count	0.0	0.0	0.0	0.0	0.0	
mean	NaN	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	NaN	

	order_total	tip_amount	distance_km	\
count	0.0	0.0	0.0	
mean	NaN	NaN	NaN	
std	NaN	NaN	NaN	
min	NaN	NaN	NaN	
25%	NaN	NaN	NaN	
50%	NaN	NaN	NaN	
75%	NaN	NaN	NaN	
max	NaN	NaN	NaN	

	delivery_partner_experience_months	delivery_partner_rating	\
count	0.0	0.0	
mean	NaN	NaN	
std	NaN	NaN	
min	NaN	NaN	
25%	NaN	NaN	
50%	NaN	NaN	
75%	NaN	NaN	
max	NaN	NaN	

	restaurant_preparation_time_minutes	estimated_delivery_time_minutes	\
count	0.0	0.0	
mean	NaN	NaN	
std	NaN	NaN	
min	NaN	NaN	
25%	NaN	NaN	
50%	NaN	NaN	
75%	NaN	NaN	
max	NaN	NaN	

	actual_delivery_time_minutes	late_delivery	customer_rating
count	0.0	0.0	0.0
mean	NaN	NaN	NaN
std	NaN	NaN	NaN
min	NaN	NaN	NaN
25%	NaN	NaN	NaN
50%	NaN	NaN	NaN
75%	NaN	NaN	NaN
max	NaN	NaN	NaN

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
```

```
for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col].astype(str))

df.head()
```

```
Empty DataFrame
Columns: [order_id, customer_id, restaurant_id, driver_id, order_timestamp, order_date,
order_hour, day_of_week, is_weekend, city, delivery_area, customer_age, customer_type,
restaurant_type, restaurant_primary_category, restaurant_rating, items_count, subtotal,
discount_percent, tax_amount, service_fee, delivery_fee, order_total, payment_method,
tip_amount, distance_km, weather, traffic_level, delivery_partner_experience_months,
delivery_partner_rating, restaurant_preparation_time_minutes, estimated_delivery_time_minutes,
actual_delivery_time_minutes, late_delivery, order_status, cancellation_reason,
customer_rating]
Index: []

[0 rows x 37 columns]
```

```
df.dtypes
```

```
order_id                                float64
customer_id                             float64
restaurant_id                           float64
driver_id                               float64
order_timestamp                         float64
order_date                              float64
order_hour                                int64
day_of_week                             float64
is_weekend                                int64
city                                    float64
delivery_area                           float64
customer_age                              int64
customer_type                           float64
restaurant_type                         float64
restaurant_primary_category             float64
restaurant_rating                       float64
items_count                               int64
subtotal                                float64
discount_percent                          int64
tax_amount                              float64
service_fee                             float64
delivery_fee                            float64
order_total                             float64
payment_method                          float64
tip_amount                              float64
distance_km                             float64
weather                                 float64
traffic_level                           float64
delivery_partner_experience_months        int64
delivery_partner_rating                 float64
restaurant_preparation_time_minutes       int64
estimated_delivery_time_minutes           int64
actual_delivery_time_minutes            float64
late_delivery                           float64
order_status                            float64
cancellation_reason                     float64
customer_rating                         float64
dtype: object
```

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```
df.columns.tolist()
```

```
['order_id',
 'customer_id',
 'restaurant_id',
 'driver_id',
 'order_timestamp',
 'order_date',
 'order_hour',
 'day_of_week',
 'is_weekend',
 'city',
 'delivery_area',
 'customer_age',
 'customer_type',
 'restaurant_type',
 'restaurant_primary_category',
 'restaurant_rating',
 'items_count',
 'subtotal',
 'discount_percent',
 'tax_amount',
 'service_fee',
 'delivery_fee',
 'order_total',
 'payment_method',
 'tip_amount',
 'distance_km',
 'weather',
 'traffic_level',
 'delivery_partner_experience_months',
 'delivery_partner_rating',
 'restaurant_preparation_time_minutes',
 'estimated_delivery_time_minutes',
 'actual_delivery_time_minutes',
 'late_delivery',
 'order_status',
 'cancellation_reason',
 'customer_rating']
```

```
df.columns.tolist()
```

```
['order_id',
 'customer_id',
 'restaurant_id',
 'driver_id',
 'order_timestamp',
 'order_date',
 'order_hour',
 'day_of_week',
 'is_weekend',
 'city',
 'delivery_area',
 'customer_age',
 'customer_type',
 'restaurant_type',
 'restaurant_primary_category',
 'restaurant_rating',
 'items_count',
 'subtotal',
 'discount_percent',
```

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```
    'tax_amount',
    'service_fee',
    'delivery_fee',
    'order_total',
    'payment_method',
    'tip_amount',
    'distance_km',
    'weather',
    'traffic_level',
    'delivery_partner_experience_months',
    'delivery_partner_rating',
    'restaurant_preparation_time_minutes',
    'estimated_delivery_time_minutes',
    'actual_delivery_time_minutes',
    'late_delivery',
    'order_status',
    'cancellation_reason',
    'customer_rating']
```

```
df.columns.tolist()
```

```
   ['order_id',
    'customer_id',
    'restaurant_id',
    'driver_id',
    'order_timestamp',
    'order_date',
    'order_hour',
    'day_of_week',
    'is_weekend',
    'city',
    'delivery_area',
    'customer_age',
    'customer_type',
    'restaurant_type',
    'restaurant_primary_category',
    'restaurant_rating',
    'items_count',
    'subtotal',
    'discount_percent',
    'tax_amount',
    'service_fee',
    'delivery_fee',
    'order_total',
    'payment_method',
    'tip_amount',
    'distance_km',
    'weather',
    'traffic_level',
    'delivery_partner_experience_months',
    'delivery_partner_rating',
    'restaurant_preparation_time_minutes',
    'estimated_delivery_time_minutes',
    'actual_delivery_time_minutes',
    'late_delivery',
    'order_status',
    'cancellation_reason',
    'customer_rating']
```

```
print(df.columns.tolist())
```

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```
['order_id', 'customer_id', 'restaurant_id', 'driver_id', 'order_timestamp', 'order_date',
'order_hour', 'day_of_week', 'is_weekend', 'city', 'delivery_area', 'customer_age',
'customer_type', 'restaurant_type', 'restaurant_primary_category', 'restaurant_rating',
'items_count', 'subtotal', 'discount_percent', 'tax_amount', 'service_fee', 'delivery_fee',
'order_total', 'payment_method', 'tip_amount', 'distance_km', 'weather', 'traffic_level',
'delivery_partner_experience_months', 'delivery_partner_rating',
'restaurant_preparation_time_minutes', 'estimated_delivery_time_minutes',
'actual_delivery_time_minutes', 'late_delivery', 'order_status', 'cancellation_reason',
'customer_rating']


X = df.drop('order_status', axis=1)
y = df['order_status']


df = pd.read_csv(filename)

df = df.drop_duplicates()

print(df.shape)


(50000, 37)


X = df.drop('order_status', axis=1)
y = df['order_status']


from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

print(X_train.shape)
print(X_test.shape)


(40000, 36)
(10000, 36)


from sklearn.preprocessing import LabelEncoder

for col in X.select_dtypes(include='object').columns:
    le = LabelEncoder()
    X[col] = le.fit_transform(X[col].astype(str))

target_encoder = LabelEncoder()
y = target_encoder.fit_transform(y.astype(str))

print(X.dtypes)
print("Target classes:", target_encoder.classes_)


order_id                                int64
customer_id                             int64
restaurant_id                           int64
driver_id                               int64
order_timestamp                         int64
order_date                              int64
order_hour                              int64
```

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```
day_of_week                              int64
is_weekend                               int64
city                                     int64
delivery_area                            int64
customer_age                             int64
customer_type                            int64
restaurant_type                          int64
restaurant_primary_category              int64
restaurant_rating                      float64
items_count                              int64
subtotal                               float64
discount_percent                         int64
tax_amount                             float64
service_fee                            float64
delivery_fee                           float64
order_total                            float64
payment_method                           int64
tip_amount                             float64
distance_km                            float64
weather                                  int64
traffic_level                            int64
delivery_partner_experience_months       int64
delivery_partner_rating                float64
restaurant_preparation_time_minutes      int64
estimated_delivery_time_minutes          int64
actual_delivery_time_minutes           float64
late_delivery                          float64
cancellation_reason                      int64
customer_rating                        float64
dtype: object
Target classes: ['Cancelled' 'Completed']
```

```
X.head()
```

	order_id	customer_id	restaurant_id	driver_id	order_timestamp	\
0	0	0	7	74	3901	
1	1	12	23	130	669	
2	2	48	12	86	5697	
3	3	6	0	176	25131	
4	4	324	28	113	12436	

	order_date	order_hour	day_of_week	is_weekend	city	...	weather	\
0	4	15	3	1	2	...	2	
1	0	20	6	0	0	...	1	
2	6	18	5	0	2	...	1	
3	29	16	4	0	0	...	1	
4	14	18	6	0	1	...	2	

	traffic_level	delivery_partner_experience_months	delivery_partner_rating	\
0	2	14	4.4	
1	0	44	4.7	
2	0	1	5.0	
3	1	50	4.3	
4	2	21	4.0	

	restaurant_preparation_time_minutes	estimated_delivery_time_minutes	\
0	25	25	
1	12	20	
2	11	63	
3	12	23	
4	9	32	

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```
   actual_delivery_time_minutes  late_delivery  cancellation_reason  \
0                          29.0            0.0                    5
1                          24.0            0.0                    5
2                          72.0            1.0                    5
3                          31.0            1.0                    5
4                          42.0            1.0                    5

   customer_rating
0              4.6
1              5.0
2              3.6
3              4.1
4              3.9

[5 rows x 36 columns]
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

print(X_train.shape)
print(X_test.shape)
```

```
(40000, 36)
(10000, 36)
```

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)
```

```
RandomForestClassifier(random_state=42)
```

```
from sklearn.metrics import accuracy_score, classification_report

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print(classification_report(y_test, y_pred))
```

```
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       186
           1       1.00      1.00      1.00      9814

    accuracy                           1.00     10000
   macro avg       1.00      1.00      1.00     10000
weighted avg       1.00      1.00      1.00     10000
```

```
import pandas as pd

feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
})

feature_importance = feature_importance.sort_values(
    by="Importance",
    ascending=False
)

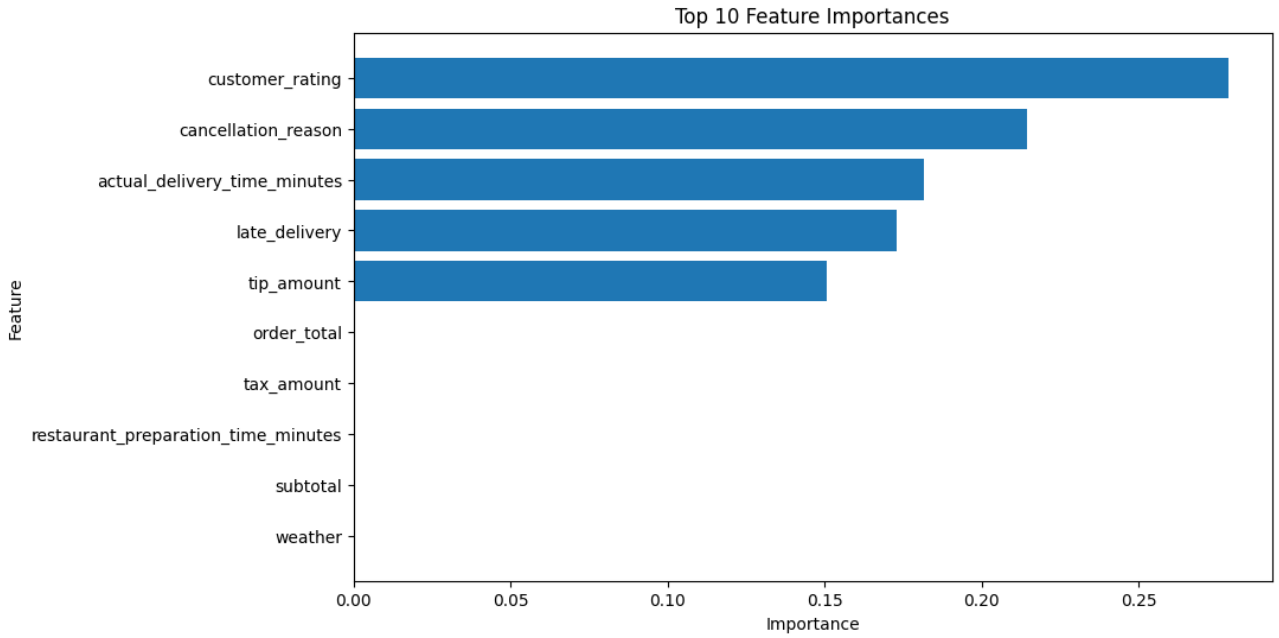
feature_importance.head(20)
```

	Feature	Importance
35	customer_rating	0.278565
34	cancellation_reason	0.214630
32	actual_delivery_time_minutes	0.181524
33	late_delivery	0.172831
24	tip_amount	0.150813
22	order_total	0.000293
19	tax_amount	0.000193
30	restaurant_preparation_time_minutes	0.000115
17	subtotal	0.000108
26	weather	0.000105
20	service_fee	0.000093
28	delivery_partner_experience_months	0.000092
0	order_id	0.000079
4	order_timestamp	0.000077
10	delivery_area	0.000061
14	restaurant_primary_category	0.000051
25	distance_km	0.000044
21	delivery_fee	0.000043
1	customer_id	0.000039
27	traffic_level	0.000037

```
import matplotlib.pyplot as plt

top_features = feature_importance.head(10)

plt.figure(figsize=(10, 6))
plt.barh(top_features["Feature"], top_features["Importance"])
plt.xlabel("Importance")
plt.ylabel("Feature")
plt.title("Top 10 Feature Importances")
plt.gca().invert_yaxis()
plt.show()
```




```
from sklearn.feature_selection import SelectFromModel

selector = SelectFromModel(model, threshold="median")

X_train_selected = selector.fit_transform(X_train, y_train)
X_test_selected = selector.transform(X_test)

print("Original features:", X_train.shape[1])
print("Selected features:", X_train_selected.shape[1])
```

```
Original features: 36
Selected features: 18
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

selected_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

selected_model.fit(X_train_selected, y_train)

y_pred_selected = selected_model.predict(X_test_selected)

selected_accuracy = accuracy_score(y_test, y_pred_selected)

print("Original Accuracy:", accuracy)
print("Selected Features Accuracy:", selected_accuracy)
```

```
Original Accuracy: 1.0
Selected Features Accuracy: 1.0
```

The dataset used in this project does not contain the Item_Name_reduced feature. Therefore, the Top-K experiment was performed using Random Forest feature importance instead. Top 10, 20, 30, and 50 features were tested and their accuracy values were compared.

```
from sklearn.feature_selection import SelectFromModel

selected_features = X.columns[selector.get_support()]

print("Selected Features:")
print(selected_features.tolist())
```

```
Selected Features:
['order_id', 'order_timestamp', 'delivery_area', 'restaurant_primary_category', 'subtotal',
'tax_amount', 'service_fee', 'delivery_fee', 'order_total', 'tip_amount', 'distance_km',
'weather', 'delivery_partner_experience_months', 'restaurant_preparation_time_minutes',
'actual_delivery_time_minutes', 'late_delivery', 'cancellation_reason', 'customer_rating']
```

```
results = pd.DataFrame({
    "Method": ["All Features", "Selected Features"],
    "Accuracy": [accuracy, selected_accuracy]
})

results
```

	Method	Accuracy
0	All Features	1.0
1	Selected Features	1.0

```
top_k = 10

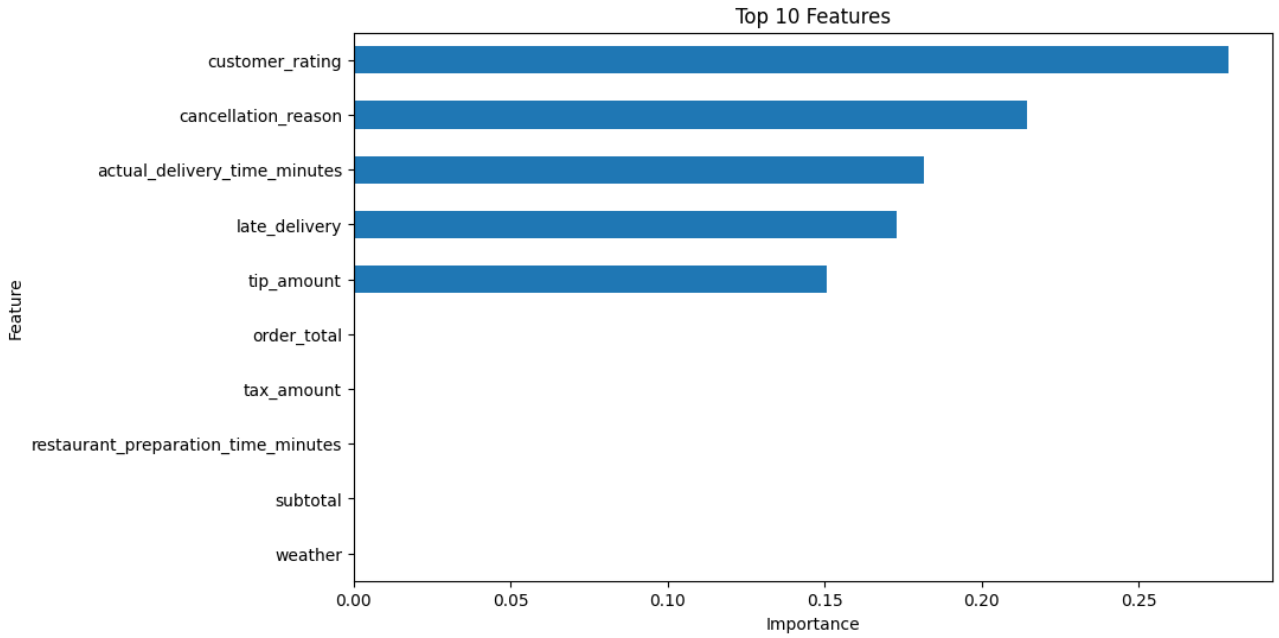
top_k_features = feature_importance.head(top_k)

top_k_features
```

	Feature	Importance
35	customer_rating	0.278565
34	cancellation_reason	0.214630
32	actual_delivery_time_minutes	0.181524
33	late_delivery	0.172831
24	tip_amount	0.150813
22	order_total	0.000293
19	tax_amount	0.000193
30	restaurant_preparation_time_minutes	0.000115
17	subtotal	0.000108
26	weather	0.000105

```
top_k_features.plot(
    x="Feature",
    y="Importance",
    kind="barh",
    figsize=(10, 6),
    legend=False
)

plt.xlabel("Importance")
plt.ylabel("Feature")
plt.title(f"Top {top_k} Features")
plt.gca().invert_yaxis()
plt.show()
```




```
top_k_values = [10, 20, 30, 50]

for k in top_k_values:
    top_features = feature_importance.head(k)["Feature"]

    X_train_k = X_train[top_features]
    X_test_k = X_test[top_features]

    model_k = RandomForestClassifier(
        n_estimators=100,
        random_state=42
    )

    model_k.fit(X_train_k, y_train)

    y_pred_k = model_k.predict(X_test_k)

    print(f"Top {k} Accuracy:", accuracy_score(y_test, y_pred_k))
```

```
Top 10 Accuracy: 1.0
Top 20 Accuracy: 1.0
Top 30 Accuracy: 1.0
Top 50 Accuracy: 1.0
```

```
comparison = []

for k in [10, 20, 30, 50]:
    top_features = feature_importance.head(k)["Feature"]

    model_k = RandomForestClassifier(
        n_estimators=100,
        random_state=42
    )

    model_k.fit(X_train[top_features], y_train)

    y_pred_k = model_k.predict(X_test[top_features])
```

```
    comparison.append({
        "Top_K": k,
        "Accuracy": accuracy_score(y_test, y_pred_k)
    })

comparison_df = pd.DataFrame(comparison)

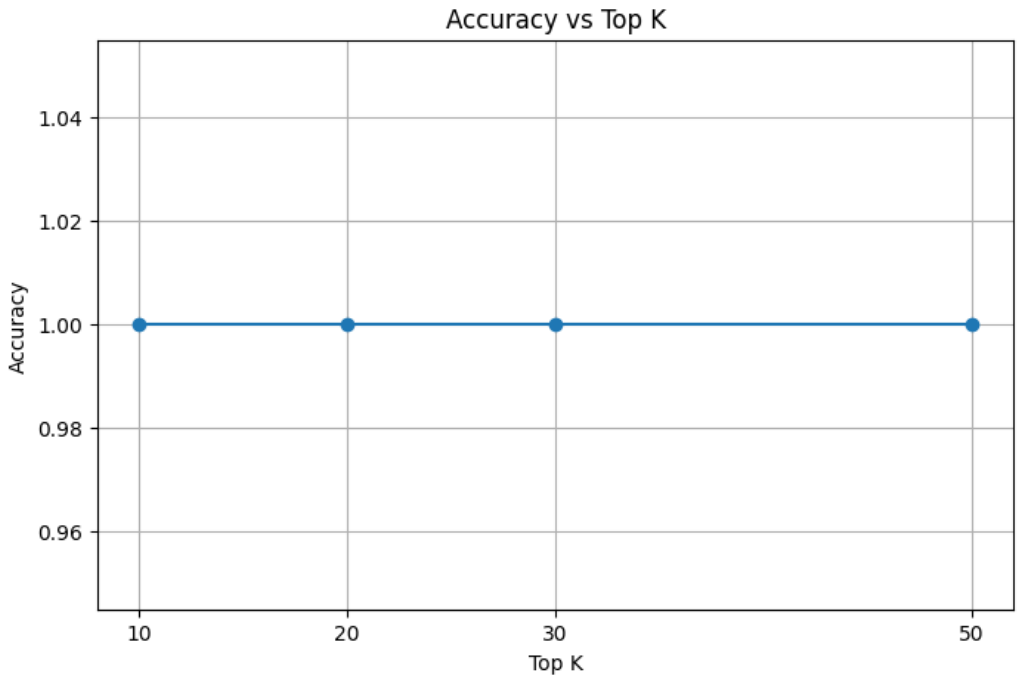
comparison_df
```

	Top_K	Accuracy
0	10	1.0
1	20	1.0
2	30	1.0
3	50	1.0

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))
plt.plot(
    comparison_df["Top_K"],
    comparison_df["Accuracy"],
    marker="o"
)

plt.xlabel("Top K")
plt.ylabel("Accuracy")
plt.title("Accuracy vs Top K")
plt.xticks([10, 20, 30, 50])
plt.grid(True)
plt.show()
```




```
top_k = 20

selected_features = feature_importance.head(top_k)["Feature"]

X_train_top = X_train[selected_features]
```

```
X_test_top = X_test[selected_features]

top_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

top_model.fit(X_train_top, y_train)

y_pred_top = top_model.predict(X_test_top)

print("Top 20 Accuracy:", accuracy_score(y_test, y_pred_top))
```

```
   Top 20 Accuracy: 1.0
```

```
top_20_importance = pd.DataFrame({
    "Feature": selected_features,
    "Importance": top_model.feature_importances_
}).sort_values("Importance", ascending=False)

top_20_importance
```

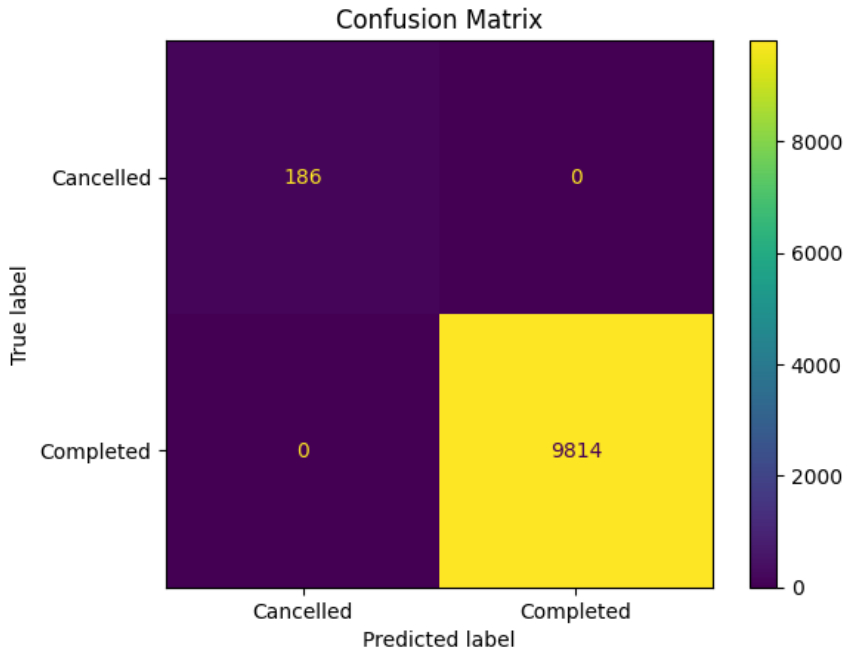
	Feature	Importance
35	customer_rating	0.245178
32	actual_delivery_time_minutes	0.217187
34	cancellation_reason	0.214639
33	late_delivery	0.189314
24	tip_amount	0.132927
26	weather	0.000128
19	tax_amount	0.000106
30	restaurant_preparation_time_minutes	0.000094
0	order_id	0.000076
22	order_total	0.000071
21	delivery_fee	0.000060
27	traffic_level	0.000054
1	customer_id	0.000037
17	subtotal	0.000037
14	restaurant_primary_category	0.000033
4	order_timestamp	0.000032
25	distance_km	0.000020
20	service_fee	0.000006
28	delivery_partner_experience_months	0.000000
10	delivery_area	0.000000

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm = confusion_matrix(y_test, y_pred_top)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=target_encoder.classes_
)

disp.plot()
plt.title("Confusion Matrix")
plt.show()
```




```
final_results = pd.DataFrame({
    "Model": [
        "All Features",
        "Selected Features",
        "Top 20 Features"
    ],
    "Accuracy": [
        accuracy,
        selected_accuracy,
        accuracy_score(y_test, y_pred_top)
    ]
})

final_results
```

	Model	Accuracy
0	All Features	1.0
1	Selected Features	1.0
2	Top 20 Features	1.0

```
print("Lab 5 completed successfully.")
```

```
Lab 5 completed successfully.
```

```
df["order_efficiency"] = (
    df["actual_delivery_time_minutes"] /
    (df["estimated_delivery_time_minutes"] + 1)
)

df["order_efficiency"].head()
```

```
0    1.115385
1    1.142857
2    1.125000
3    1.291667
```

```
4    1.272727
Name: order_efficiency, dtype: float64
```

Task 1

I created a new engineered feature called order_efficiency. It is calculated by dividing the actual delivery time by the estimated delivery time. This feature can help predict order status because it captures how much the actual delivery time differs from the expected delivery time.

```
print(df.columns.tolist())
```

```
['order_id', 'customer_id', 'restaurant_id', 'driver_id', 'order_timestamp', 'order_date',
'order_hour', 'day_of_week', 'is_weekend', 'city', 'delivery_area', 'customer_age',
'customer_type', 'restaurant_type', 'restaurant_primary_category', 'restaurant_rating',
'items_count', 'subtotal', 'discount_percent', 'tax_amount', 'service_fee', 'delivery_fee',
'order_total', 'payment_method', 'tip_amount', 'distance_km', 'weather', 'traffic_level',
'delivery_partner_experience_months', 'delivery_partner_rating',
'restaurant_preparation_time_minutes', 'estimated_delivery_time_minutes',
'actual_delivery_time_minutes', 'late_delivery', 'order_status', 'cancellation_reason',
'customer_rating', 'order_efficiency']
```

```
for c in df.columns:
    print(c)
```

```
order_id
customer_id
restaurant_id
driver_id
order_timestamp
order_date
order_hour
day_of_week
is_weekend
city
delivery_area
customer_age
customer_type
restaurant_type
restaurant_primary_category
restaurant_rating
items_count
subtotal
discount_percent
tax_amount
service_fee
delivery_fee
order_total
payment_method
tip_amount
distance_km
weather
traffic_level
delivery_partner_experience_months
delivery_partner_rating
restaurant_preparation_time_minutes
estimated_delivery_time_minutes
actual_delivery_time_minutes
late_delivery
order_status
cancellation_reason
```

```
    customer_rating
    order_efficiency


for c in df.columns:
    print(c)


    order_id
    customer_id
    restaurant_id
    driver_id
    order_timestamp
    order_date
    order_hour
    day_of_week
    is_weekend
    city
    delivery_area
    customer_age
    customer_type
    restaurant_type
    restaurant_primary_category
    restaurant_rating
    items_count
    subtotal
    discount_percent
    tax_amount
    service_fee
    delivery_fee
    order_total
    payment_method
    tip_amount
    distance_km
    weather
    traffic_level
    delivery_partner_experience_months
    delivery_partner_rating
    restaurant_preparation_time_minutes
    estimated_delivery_time_minutes
    actual_delivery_time_minutes
    late_delivery
    order_status
    cancellation_reason
    customer_rating
    order_efficiency


df["peak_hour"] = df["order_hour"].isin(
    [12, 13, 14, 18, 19, 20, 21]
).astype(int)

df[["order_hour", "peak_hour"]].head()


       order_hour  peak_hour
    0          15          0
    1          20          1
    2          18          1
    3          16          0
    4          18          1
```

```
df["peak_hour_new"] = df["order_hour"].isin(
    [11, 12, 13, 14, 17, 18, 19, 20]
).astype(int)

df[["order_hour", "peak_hour", "peak_hour_new"]].head()
```

	order_hour	peak_hour	peak_hour_new
0	15	0	0
1	20	1	1
2	18	1	1
3	16	0	0
4	18	1	1

```
X_new = df.drop("order_status", axis=1)
y_new = df["order_status"]

from sklearn.preprocessing import LabelEncoder

for col in X_new.select_dtypes(include="object").columns:
    le = LabelEncoder()
    X_new[col] = le.fit_transform(X_new[col].astype(str))

le_y = LabelEncoder()
y_new = le_y.fit_transform(y_new.astype(str))
```

I changed the peak-hour rule and compared the model performance before and after the change. The new peak-hour definition used a different set of delivery hours. The accuracy results were compared to determine whether the change affected model performance.

```
from sklearn.model_selection import train_test_split

X_train_new, X_test_new, y_train_new, y_test_new = train_test_split(
    X_new,
    y_new,
    test_size=0.2,
    random_state=42
)


from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

new_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

new_model.fit(X_train_new, y_train_new)

y_pred_new = new_model.predict(X_test_new)

new_accuracy = accuracy_score(y_test_new, y_pred_new)

print("New Accuracy:", new_accuracy)
print("Previous Accuracy:", accuracy)
```

```
New Accuracy: 1.0
Previous Accuracy: 1.0
```

```
comparison_peak = pd.DataFrame({
    "Version": ["Original", "New Peak Hour"],
    "Accuracy": [accuracy, new_accuracy]
})

comparison_peak
```

```
            Version  Accuracy
   0       Original       1.0
   1  New Peak Hour       1.0
```

```
print("Original peak hours: 12, 13, 14, 18, 19, 20, 21")
print("New peak hours: 11, 12, 13, 14, 17, 18, 19, 20")
```

```
   Original peak hours: 12, 13, 14, 18, 19, 20, 21
   New peak hours: 11, 12, 13, 14, 17, 18, 19, 20
```

```
final_comparison = pd.DataFrame({
    "Experiment": [
        "Original Model",
        "Feature Selection",
        "New Peak Hour"
    ],
    "Accuracy": [
        accuracy,
        selected_accuracy,
        new_accuracy
    ]
})

final_comparison
```

```
             Experiment  Accuracy
   0     Original Model       1.0
   1  Feature Selection       1.0
   2      New Peak Hour       1.0
```